On a statement of A. N. Shoshitaishvili about weighted homogeneous singularities

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Abstract

Let F be a weighted homogeneous polynomial with an isolated critical point at 0, with weights $w_1 \geq \cdots \geq w_n$ and degree d. Statement 1 of the author's 1976 paper says that if $\varphi(0) = 0$ and the Jacobian ideal of φ lies in the Jacobian ideal of F, then φ lies in the Jacobian ideal of F and has weighted degree at least d. The second part is equivalent to the absence of nonzero derivations of negative weight of the Milnor algebra of F; since 1995 Shoshitaishvili's Statement 1 has been studied as a conjecture of S. S.-T. Yau, and as the Milnor-algebra case of Halperin's conjecture. We describe the history of the statement and give a proof of the following: with $M = \min(w_3 - 2w_n,\ w_1 - 3w_n)$, the Milnor algebra has no nonzero derivation of weight $-q$ if $q > M$, or if $q > M - (w_{n-1} - w_n)$ and w_n does not divide $d - q$; in particular it has no nonzero derivation of negative weight when $w_3 \leq 2w_n$ or $w_1 \leq 3w_n$.

1 Introduction

Notation. $P = \mathbb{C}[x_1, \ldots, x_n]$, with positive integer weights $w_1 \geq w_2 \geq \cdots \geq w_n$ of the variables. A polynomial is weighted homogeneous of weight e if all its monomials $x_1^{c_1} \cdots x_n^{c_n}$ have $\sum_i c_i w_i = e$. $F \in P$ is weighted homogeneous of degree d and has an isolated critical point at 0. Throughout the paper weights and degrees are assumed to be integers, and $\gcd(w_1, \ldots, w_n) = 1$; since $F \in \mathfrak{m}^3$ (see the reduction to $d > 2w_1$ below), this makes $(w_1, \ldots, w_n, d)$ unique for F [Sa71]. There are two exceptions: Statement 1 as quoted from [S76] below, and its discussion in Section 2.1. There, as in [S76], weights are normalized: $\alpha_i = w_i/d$ (rational; $0 < \alpha_i < 1/2$ because $d > 2w_1$, while [S76] allows $\alpha_i \leq 1/2$), F has degree 1, and a weighted homogeneous polynomial of weight e has normalized weight e/d. $F_i = \partial F/\partial x_i$, $J = J_F = (F_1, \ldots, F_n)$, $\nabla F = (F_1, \ldots, F_n)^T$, and $\mathrm{Mil}(F) = P/J$ is the Milnor algebra, graded by weight. For any polynomial φ, J_φ is the ideal generated by its partial derivatives. $\mathfrak{m} = (x_1, \ldots, x_n)$.

Grading. $P = \bigoplus_e P_e$, where P_e is the finite-dimensional space of weighted homogeneous polynomials of weight e. Since J is generated by weighted homogeneous polynomials, $J = \bigoplus_e (J \cap P_e)$: a polynomial lies in J if and only if each of its weighted homogeneous components does. So every membership in J and every relation among $F_1, \ldots, F_n$ can be checked separately in each weight. In particular, in an identity $A = \sum_j c_j F_j$ with A weighted homogeneous of weight e, each c_j can be replaced by its component of weight $e - (d - w_j)$. Likewise, a vector field D with $D(J) \subset J$ is

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a sum of weighted homogeneous vector fields, and each of them maps J into J. So it is enough to consider derivations of a single weight $-q$.

We assume $d > 2w_1$. This is no loss: by K. Saito [Sa71], after a coordinate change $F = x_1^2 + \cdots + x_k^2 + G$, where $G \in \mathfrak{m}^3$ is in the other variables, which keep their weights, and $\mathrm{Mil}(F) \cong \mathrm{Mil}(G)$ as graded algebras. (This assumption excludes any quadratic terms in F; for example, $F = xy + z^3$ is not possible.) With $d > 2w_1$, every monomial of ordinary degree ≤ 2 has weight $\le 2w_1 < d$, so $F \in \mathfrak{m}^3$.

A derivation of $\mathrm{Mil}(F)$ of weight $-q$, $q > 0$, is induced by a vector field $D = \sum_i p_i \partial_i$ with $p_i \in P$ weighted homogeneous of weight $w_i - q$ and $D(J) \subset J$. The least weight of a nonzero element of J is $d - w_1$, and $w_i - q < w_1 < d - w_1$, so the p_i are determined by the derivation, and the derivation is 0 if and only if all $p_i = 0$.

Statement 1 of [S76]. *Let $F \in J$, $d_\omega F = F$, $\dim_{\mathbb{C}} Q(F) = \mu < \infty$, $\varphi \in M \subset \mathbb{C}[[x_1, \dots, x_n]]$, $i(\varphi) \subset i(F)$. Then $\varphi \in i(F)$ and $\lambda(\varphi) \ge 1$.*

J is the set of germs F of holomorphic functions at $0 \in \mathbb{C}^n$ with $F(0) = 0$, $\operatorname{grad} F(0) = 0$ (in the rest of this paper J denotes the Jacobian ideal). d_ω is differentiation along the vector field $\omega = \sum_i \alpha_i x_i \partial_{x_i}$ with rational $0 < \alpha_i \le 1/2$; so $d_\omega F = F$ says that F has degree 1 for the normalized weights $\alpha_i = w_i/d$ (see Notation). $i(F) \subset \mathbb{C}[[x_1, \dots, x_n]]$ is the ideal generated by the partial derivatives of F, i.e. J_F, and $Q(F) = \mathbb{C}[[x_1, \dots, x_n]]/i(F)$; so $\mu < \infty$ says that the critical point 0 is isolated. M is the maximal ideal of $\mathbb{C}[[x_1, \dots, x_n]]$; so $\varphi \in M$ means $\varphi(0) = 0$. $\lambda(\varphi)$ is the normalized weight of φ: if φ is weighted homogeneous of weight e, then $d_\omega \varphi = (e/d)\varphi$ and $\lambda(\varphi) = e/d$; in general $\lambda(\varphi) = e/d$, where e is the weight of the lowest weighted homogeneous part of φ. So $\lambda(\varphi) \ge 1$ says that this lowest part has weight $\ge d$, the degree of F. Since $i(F)$ is generated by weighted homogeneous polynomials, membership in $i(F)$ can be checked on each weighted homogeneous part (see Grading), so for a polynomial φ the conclusion $\varphi \in i(F)$ is $\varphi \in J_F$.

The first part is proved in [S76]. For each F, the second part is equivalent to: $\mathrm{Mil}(F)$ has no nonzero derivation of negative weight (Section 2.1). Section 2 describes the history of the statement. Our result is the following.

Theorem 1.1. *Let F be as above, $n \ge 3$, $q \ge 1$, and*

$$M = \min(w_3 - 2w_n,\ w_1 - 3w_n).$$

If $q > M$, or if $q > M - (w_{n-1} - w_n)$ and w_n does not divide $d - q$, then $\mathrm{Mil}(F)$ has no nonzero derivation of weight $-q$. In particular, if $w_3 \le 2w_n$ or $w_1 \le 3w_n$, then $\mathrm{Mil}(F)$ has no nonzero derivation of negative weight, and the second part of Statement 1 holds for F.

Theorem 1.1 is new, and its proof is independent of the published proofs. Among the published results whose proofs are available, it is stronger than every condition on the weights alone that applies for all n in the papers listed in Section 2.4: it strictly contains the condition $w_1 \le 2w_n$ of [YZ16] (whose proof needs the repair described in Section 2.6) and the case of equal weights. It does not contain, and is not contained in, the condition $d > w_1^2 - w_1 + 1$ of [CY26], which uses the degree d (Examples 7.1 and 7.2). The announced general proof [Al26] is not part of this comparison, because its proof is not available (Section 2.5).

Theorem 1.1 follows from a more flexible criterion, Theorem 6.1: for each row of the matrix C of Section 3 one of three conditions on the weights rules out that this row is the first nonzero row of C, and different rows may be excluded by different conditions (Example 7.3).

2 History of the statement

2.1 Statement 1 and the paper [S76]

Statement 1 appears in the author's paper [S76] of 1976. The first part, $\varphi \in J_F$, is Lemma 2 of [S76] (pp. 59–60), and its proof is correct: J_F is generated by weighted homogeneous polynomials, so each weighted homogeneous part φ_e of φ (of weight e; here $e > 0$, because $\varphi \in M$ gives $\varphi(0) = 0$) also satisfies $J_{\varphi_e} \subset J_F$, and Euler's identity $\varphi_e = e^{-1} \sum_k w_k x_k \, \partial_k \varphi_e$ puts φ_e in J_F.

For the second part [S76] gives a residue argument (p. 61), and there is a gap in that argument. Let φ be weighted homogeneous of normalized weight $\lambda < 1$ (see Notation); here $\lambda > 0$, because $\varphi \in M$ gives $\varphi(0) = 0$. The argument substitutes $x_i = t^{\alpha_i} x_i'$ in the residue integral

$$I = \int \frac{\det(\partial_i \partial_j F)\, dx}{\prod_i (\partial_i \varphi + F_i)}$$

and writes, in formula (5) of [S76], $I = t^{k(1-\lambda)} \Pi_t$, where k is the number of i with $\partial_i \varphi \neq 0$ and Π_t is the same integral with denominators $\partial_i \varphi + t^{1-\lambda} F_i$. Comparing the two sides it concludes $k = n$, which contradicts $\partial_1 \varphi = 0$ (the least weight in J_F is $d - w_1$). But the substitution multiplies every factor $\partial_i \varphi + F_i$, also those with $\partial_i \varphi = 0$, by $t^{\lambda - \alpha_i}$, so the correct factor is $t^{n(1-\lambda)}$, not $t^{k(1-\lambda)}$. With the correct factor, formula (5) is the residue identity for $\varphi + t^{1-\lambda} F$, which holds for every φ, and $k = n$ does not follow. So the second part is not proved in [S76].

The second part in terms of derivations. For each F the second part is equivalent to: $\mathrm{Mil}(F)$ has no nonzero derivation of negative weight. If $J_\varphi \subset J_F$ with φ of weight $< d$ (and of weight > 0, because $\varphi \in M$ gives $\varphi(0) = 0$), then $\varphi = D(F)$ for a vector field D of negative weight with $D(J_F) \subset J_F$. Indeed, $\varphi \in J_F$ by quasihomogeneity (Euler's identity, as in the first part), so $\varphi = \sum_j p_j \partial_j F = D(F)$ with $D = \sum_j p_j \partial_j$. Conversely $D(F)$ satisfies $J_{D(F)} \subset J_F$, and $D \neq 0$ gives $D(F) \neq 0$ by Lemma 3.1.

2.2 Halperin's conjecture

The first publication of S. Halperin's conjecture in rational homotopy theory was in 1977 [Ha77]: for every orientable fibration whose fibre X is an elliptic space with cohomology only in even degrees, the rational Serre spectral sequence collapses at E_2. For such X, $H^*(X;\mathbb{Q})$ is a graded Artinian complete intersection $\mathbb{Q}[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ ([Ha77], 1977). In 1982 W. Meier [Me82, Theorem A] showed that the collapse for every orientable fibration with fibre X is equivalent to $H^*(X;\mathbb{Q})$ having no nonzero derivation of negative degree. Since 1982 Halperin's conjecture has been stated in this algebraic form: a graded Artinian complete intersection has no nonzero derivation of negative weight ([CY26, Conjecture 1.1], 2026). One has: $\mathrm{Mil}(F) = P/(\partial F/\partial x_1, \ldots, \partial F/\partial x_n)$ is a graded Artinian complete intersection. So the second part of Statement 1 would follow from Halperin's conjecture for general graded Artinian complete intersections.

2.3 The name used since 1995

The case of Milnor algebras was proposed independently by S. S.-T. Yau, and in the literature since 1995 it has been called Yau's conjecture ([CXY95], [Ch95]; [CY26], Conjecture 1.2). It is the second part of Statement 1 of [S76]. In these terms it was proved for $n = 2, 3$ by H. Chen, Y.-J. Xu and Yau [CXY95], for $n = 4$ by H. Chen [Ch95], for $w_n \geq w_1/2$ by Yau and H. Zuo [YZ16], and for $d > w_1^2 - w_1 + 1$ by B. Chen and Yau [CY26]. The proof in [YZ16] has a gap (Section 2.6).

2.4 Published work

The tables list the published papers on the second part of Shoshitaishvili's Statement 1, in the form for Milnor algebras and in the general form for Artinian complete intersections (Halperin's conjecture), which contains it as the case $f_i = \partial F/\partial x_i$. The lists are compiled from the bibliographies of [CY26] and [CYZ19] and the papers cited there.

Milnor algebras.

authors	year	content
A. N. Shoshitaishvili [S76]	1976	Statement 1: $\varphi(0) = 0$ and $J_\varphi \subset J_F$ imply $\varphi \in J_F$ (proved) and weighted degree $\geq d$ (residue argument with a gap)
H. Chen, Y.-J. Xu, S. S.-T. Yau [CXY95]	1995	the statement in terms of derivations; proof for $n = 2, 3$
H. Chen [Ch95]	1995	proof for $n = 4$
Y.-J. Xu, S. S.-T. Yau [XY96]	1996	micro-local characterization of quasi-homogeneous singularities by the derivation Lie algebra of $\mathrm{Mil}(F)$, assuming the statement
S. S.-T. Yau, H. Zuo [YZ16]	2016	$w_n \geq w_1/2$; the proof has a gap (Section 2.6)
B. Chen, S. S.-T. Yau [CY26]	2026	$d > w_1^2 - w_1 + 1$

Artinian complete intersections: Halperin's conjecture.

authors	year	content
S. Halperin [Ha77], [FHT01]	1977	the conjecture (collapse of the rational Serre spectral sequence), first published in [Ha77]
J.-C. Thomas [Th81]	1981	$n = 2$
W. Meier [Me82]	1982	equivalence with the absence of negative derivations; homogeneous fibres
H. Shiga, M. Tezuka [ST87]	1987	cohomology of homogeneous spaces G/U of equal rank
G. Lupton [Lu90]	1990	$n = 3$
M. Markl [Mar90]	1990	relations homogeneous in the ordinary degree
A. G. Aleksandrov, B. Martin [AM92]	1992	equal weights; $n = 2$
S. Papadima, L. Paunescu [PP96]	1996	$P/(f_2, \ldots, f_n)$ reduced and $\deg f_1 \geq \deg f_i$
H. Chen [Ch99]	1999	$n = 3$
H. Chen, S. S.-T. Yau, H. Zuo [CYZ19]	2019	graded Artinian algebras (not necessarily complete intersections) whose relations have large degrees
L. Kennard, Y. Wu [KW23]	2023	formal dimension ≤ 20
A. G. Aleksandrov [Al26]	2026	announces a proof of the conjecture in general (Section 2.5)

Conjectures of the same type have been proposed for other algebras attached to F: the k-th Hessian algebras [MYZ23] and the higher Nash blow-up local algebras [HMYZ23].

2.5 The announced general proof

A. G. Aleksandrov, *Derivations of Artinian complete intersections* [Al26] (Algebra i Analiz 38:1 (2026), in Russian; received 9 June 2025), states in its abstract: "The main result is a proof of a long-standing conjecture which asserts that there are no derivations of negative weights on positively graded Artinian complete intersections." The same result is announced in A. G. Aleksandrov, *Vector fields on zero-dimensional complete intersections* [Al25], and was conjectured in [Al91]. Both abstracts name as the key step that the canonical primary decomposition of an ideal invariant under a derivation is itself invariant. For weighted homogeneous F with an isolated critical point,

$\mathrm{Mil}(F)$ is a positively graded Artinian complete intersection, so the announced theorem contains the second part of Statement 1 for every such F.

The full texts of [Al26] and [Al25] are not openly available, and we could not read the proof. B. Chen and Yau [CY26], published in 2026, list the general case as open. The results of the present paper do not use [Al26] or [Al25].

2.6 The gap in [YZ16]

Let D be a vector field of weight $-q < 0$ with $D(J_F) \subset J_F$. Lemma 3.1 of [YZ16] gives, for $i \geq 2$,

$$\partial_i(DF) = q_1 F_1 + \cdots + q_{i-1} F_{i-1},$$

with q_j weighted homogeneous of weight $w_j - w_i - q$ and depending only on $x_{j+1}, \ldots, x_n$. The proof of the Main Theorem of [YZ16] (p. 24) then says: "If $q(x_2, x_3, \ldots, x_n)$ is not 0, then $\deg q(x_2, x_3, \ldots, x_n) \geq \mathrm{wt}(x_n) = w_n$", and concludes that all $q_j = 0$. A nonzero constant q_j has weight 0, not $\geq w_n$, and the weights allow a nonzero constant q_j exactly when $w_j = w_i + q$. Excluding constants needs the isolatedness of the critical point, which this step does not use. The step fails for a non-isolated F that satisfies the other hypotheses: $F = zy^3$ with $\mathrm{wt}(z) = 2$, $\mathrm{wt}(y) = 1$, $d = 5$, and $D = y\,\partial_z$ (weight -1) has $D(J_F) \subset J_F$ and $\partial_y(DF) = 4y^3 = 4\,\partial F/\partial z$, so $q_1 = 4 \neq 0$, while $D \neq 0$. For isolated F the constant case is settled by Lemma 4.1(a) below, and the Main Theorem of [YZ16] ($w_1 \leq 2w_n$) is contained in Theorem 1.1 ($w_1 \leq 3w_n$).

3 The matrix C and its first nonzero row

Lemma 3.1. *Let $q > 0$ be arbitrary. If D induces a derivation of weight $-q < 0$ and $D(F) = 0$, then $D = 0$.*

Proof. $D(F) = \sum_i p_i F_i = 0$ is a relation among $F_1, \ldots, F_n$, which form a regular sequence (the critical point is isolated). The relations of a regular sequence are generated by the trivial ones $F_j e_i - F_i e_j$, so every p_i lies in J. But p_i has weight $w_i - q < d - w_1$, the least weight of a nonzero element of J. So $p_i = 0$. □

The matrix C. Let $D = \sum_i p_i \partial_i$ induce a derivation of weight $-q < 0$, and $g = D(F)$. Since $D(F_i) \in J$, there is a matrix B of weighted homogeneous polynomials with $D(\nabla F) = B \nabla F$, B_{ij} of weight $w_j - w_i - q$. Let $P_D = (\partial p_i/\partial x_j)$. From $g = \sum_k p_k F_k$,

$$\partial_i g = \sum_k (\partial_i p_k) F_k + \sum_k p_k\, \partial_k F_i = (P_D^T \nabla F)_i + (D(\nabla F))_i,$$

so

$$\nabla g = C\, \nabla F, \qquad C = B + P_D^T,$$

an identity of polynomials, with C_{ij} weighted homogeneous of weight $w_j - w_i - q$. $C_{ij} \neq 0$ needs $w_j \geq w_i + q > w_i$, hence $j < i$: C is strictly lower triangular.

If $C = 0$, then $\nabla g = 0$, g is a constant of weight $d - q > 0$, so $g = 0$ and $D = 0$ by Lemma 3.1. So to prove that there is no nonzero derivation of weight $-q$ it is enough to show $C = 0$.

The first nonzero row. Suppose $C \neq 0$, and let k be its first nonzero row. Rows $1, \dots, k-1$ vanish, so $\partial_1 g = \cdots = \partial_{k-1} g = 0$: g depends only on $x_k, \dots, x_n$. Put

$$V = \sum_{j<k} C_{kj}\, \partial_j, \qquad h = V(F) = \sum_{j<k} C_{kj} F_j = \partial_k g.$$

Since $h = \partial_k g$ is the k-th row of $\nabla g = C \nabla F$ and g depends only on $x_k, \dots, x_n$, h also depends only on $x_k, \dots, x_n$. $V(x_j) = C_{kj}$ for $j < k$, and $V(x_i) = 0$ for $i \geq k$. We call x_j *moved* if $V(x_j) \neq 0$ and *fixed* otherwise. In particular $x_k, \dots, x_n$ are fixed, so $V(h) = 0$. Each term $C_{kj}\partial_j$ has weight $(w_j - w_k - q) - w_j = -(w_k + q) < 0$, so V lowers weight and is locally nilpotent, i.e. nilpotent in application to any particular polynomial. C_{kj} has weight $w_j - w_k - q < w_j$, so it does not involve x_j or any variable of weight $\geq w_j$.

4 Lemmas on the first nonzero row

Lemma 4.1. *In the setting above:*

(a) every entry of row k of C that is a constant is 0;

(b) row k has at least two nonzero entries.

Proof. (a) Let $\Phi_\tau = \exp(\tau V)$, a polynomial automorphism of $\mathbb{C}^n$ for each $\tau \in \mathbb{C}$. Since for $h = V(F)$ one has $V(h) = 0$,

$$F \circ \Phi_\tau = F + \tau h.$$

$F \in \mathfrak{m}^3$ gives $F_j \in \mathfrak{m}^2$, so $h \in \mathfrak{m}^2$, and 0 is a critical point of $F + \tau h$. By the chain rule, $0 = D\Phi_\tau(0)^T\, \nabla F(\Phi_\tau(0))$ with $D\Phi_\tau(0)$ invertible (indeed, $\Phi_{-\tau} \circ \Phi_\tau$ is the identity, so by the chain rule $D\Phi_{-\tau}(\Phi_\tau(0))\, D\Phi_\tau(0) = I$), so $\Phi_\tau(0)$ is a critical point of F. The critical set of F is invariant under $x_i \mapsto t^{w_i} x_i$, so the isolated critical point 0 is the only one. Hence $\Phi_\tau(0) = 0$ for all τ, and $V(0) = \frac{d}{d\tau}\Phi_\tau(0)|_{\tau=0} = 0$, i.e. $C_{kj}(0) = 0$ for all j. A weighted homogeneous constant C_{kj} equals $C_{kj}(0)$, so it is 0.

(b) Suppose row k has only one nonzero entry C_{kj}, $j < k$. Then $h = C_{kj} F_j$ is a single term. h does not contain x_j, since it depends only on $x_k, \dots, x_n$, and C_{kj} does not contain x_j, since its weight is $< w_j$. With one term there is nothing to cancel the x_j in F_j, so F_j does not contain x_j: $0 = \partial_j h = C_{kj}\, \partial_j^2 F$, hence $\partial_j^2 F = 0$. Then $F = A + x_j B$ with A, B free of x_j, $A \in \mathfrak{m}^3$, $B \in \mathfrak{m}^2$, and the whole x_j-axis is critical (there $\partial_j F = B(0) = 0$ and $\partial_i F = \partial_i A(0) + x_j\, \partial_i B(0) = 0$), which contradicts the isolated critical point. □

Lemma 4.2. *Let $F \in \mathbb{C}[u, v, z_1, \dots, z_s]$ lie in $\mathfrak{m}^3$ and have an isolated critical point at 0. Let $a \in \mathbb{C}[v, z]$, $b \in \mathbb{C}[z]$, $V = a\, \partial_u + b\, \partial_v$, and suppose $V(F) \in \mathbb{C}[z]$. Then $a = b = 0$.*

No weights are used here.

Proof. Suppose $V \neq 0$ and put $h = V(F) \in \mathbb{C}[z]$.

Step 1: common factors. Let $e = \gcd(a, b)$ in $\mathbb{C}[v, z]$ ($e = a$ if $b = 0$; if $b \neq 0$, e divides b, so $e \in \mathbb{C}[z]$). Write $a = ea'$, $b = eb'$, $V' = a'\partial_u + b'\partial_v$. Then $e\, V'(F) = h$. If $h \neq 0$: when two nonzero polynomials are multiplied, the terms with the highest powers of u and v multiply and cannot cancel; h contains no u or v, so neither does $V'(F)$, i.e. $V'(F) \in \mathbb{C}[z]$. If $h = 0$, then $V'(F) = 0$. So V' satisfies the hypotheses, with $\gcd(a', b') = 1$ (and $a' = 1$ if $b' = 0$).

Step 2: $V'(0) = 0$. V' is locally nilpotent, so $\Phi_\tau = \exp(\tau V')$ is a polynomial automorphism and $F \circ \Phi_\tau = F + \tau h'$ with $h' = V'(F) \in \mathbb{C}[z] \cap \mathfrak{m}^2$. As in Lemma 4.1(a), $\Phi_\tau(0)$ is a critical point of F

for every τ, depends polynomially on τ and equals 0 at $\tau = 0$; so $\Phi_\tau(0) = 0$ and $(a'(0), b'(0)) = 0$. In particular $a' \neq 1$, so $b' \neq 0$ and $b'(0) = 0$.

From now on write a, b for a', b': $\gcd(a, b) = 1$, $b \in \mathbb{C}[z]$, $b \neq 0$, $b(0) = 0$, $V(F) = h \in \mathbb{C}[z]$.

Step 3: $F_{uu} = bK$, $F_{uv} = -aK$. ∂_u commutes with V and $\partial_u h = 0$, so $aF_{uu} + bF_{uv} = 0$. As a, b are coprime in $\mathbb{C}[u, v, z]$, b divides F_{uu}, so $F_{uu} = bK$ for a polynomial K, and then $F_{uv} = -aK$.

Step 4: every prime factor of b divides K. Let $\pi \in \mathbb{C}[z]$ be an irreducible factor of b. Then $R = \mathbb{C}[u, v, z]/(\pi) = (\mathbb{C}[z]/(\pi))[u, v]$ is a domain and $\pi \nmid a$. Modulo π: $F_{uu} \equiv 0$, so the class $\bar{F}_u$ does not involve u; and $V(F) \equiv \bar{a}\bar{F}_u$, so $\bar{a}\,\bar{F}_u = \bar{h}$ with $\bar{a} \neq 0$ and $\bar{h} \in \mathbb{C}[z]/(\pi)$. In $(\mathbb{C}[z]/(\pi))[v]$ the product of two nonzero elements is nonzero, so if $\bar{F}_u$ contained v, the highest power of v in $\bar{a}$ times the highest power in $\bar{F}_u$ would appear in the product, and $\bar{h}$ would contain v. Hence $\bar{F}_u$ contains no v (if $\bar{h} = 0$, then $\bar{F}_u = 0$ for the same reason). So $\partial_v \bar{F}_u = 0$; since π does not involve v, this means $\partial_v F_u = F_{uv} = -aK \equiv 0 \pmod{\pi}$, and $\pi \mid K$ because π does not divide a.

Step 5. By Step 4, $\mathrm{rad}(b) \mid K$, so $F_{uu} = b\,\mathrm{rad}(b)L$ and $F_{uv} = -a\,\mathrm{rad}(b)L$ for a polynomial L. Since $b(0) = 0$, b and $\mathrm{rad}(b)$ lie in $(z_1, \ldots, z_s)$. Hence $F_{uu} \in (z)^2$ and $F_{uv} \in (z)$.

Step 6: the u-axis is critical. On the u-axis ($v = 0$, $z = 0$): $F_{uu} = 0$, so $F_u = F_u(0) = 0$; $F_{uv} = 0$, so $F_v = F_v(0) = 0$; and $\partial_{z_i} F_{uu} \in (z)$, so $F_{uuz_i} = 0$ there and $F_{z_i}(u, 0, 0) = F_{z_i}(0) + u\,F_{uz_i}(0) = 0$, since $F \in \mathfrak{m}^3$. So ∇F vanishes on the u-axis, contradicting isolatedness. □

Corollary 4.3. *The first nonzero row of C has at least three nonzero entries, and all of them are nonconstant.*

Proof. By Lemma 4.1 the nonzero entries are nonconstant and there are at least two. Suppose there are exactly two, C_{kj_1}, C_{kj_2} with $j_1 < j_2 < k$. Put $u = x_{j_1}$, $v = x_{j_2}$, and let z be the other variables. C_{kj_2} has weight $< w_{j_2} \leq w_{j_1}$, so it involves neither u nor v; C_{kj_1} does not involve u. $V = C_{kj_1}\partial_u + C_{kj_2}\partial_v$ has $V(F) = h \in \mathbb{C}[z]$. Lemma 4.2 gives $V = 0$, a contradiction. □

5 Linear vector fields

Lemma 5.1. *Let R be a commutative ring containing $\frac{1}{2}$, $a = (a_1, \ldots, a_m)^T$ with $a_1, \ldots, a_m$ a regular sequence in R, $I = (a_1, \ldots, a_m)$, and H a symmetric $m \times m$ matrix over R with $Ha = 0$. Then every entry of H lies in I^2.*

Proof. For a regular sequence, I/I^2 is a free R/I-module with basis $\bar{a}_1, \ldots, \bar{a}_m$, and I^2/I^3 is a free R/I-module with basis $\bar{a}_j\bar{a}_l$ ($j \leq l$) ([Mat86], §16: a regular sequence is quasi-regular). From $\sum_j H_{ij}a_j = 0$ and freeness of I/I^2, $H_{ij} \in I$; write $H_{ij} = \sum_l T_{ijl}a_l$. Symmetry of H gives $\sum_l (T_{ijl} - T_{jil})a_l = 0$, so $T_{ijl} \equiv T_{jil} \pmod{I}$. Next, $\sum_{j,l} T_{ijl}a_j a_l = 0$, so in I^2/I^3 the coefficient of $\bar{a}_j\bar{a}_l$ vanishes: $\bar{T}_{ijl} = -\bar{T}_{ilj}$ (using $\frac{1}{2}$ for $j = l$). So $\bar{T}$ is symmetric in the first two indices and antisymmetric in the last two, and

$$\bar{T}_{ijl} = \bar{T}_{jil} = -\bar{T}_{jli} = -\bar{T}_{lji} = \bar{T}_{lij} = \bar{T}_{ilj} = -\bar{T}_{ijl},$$

so $\bar{T} = 0$, $T_{ijl} \in I$ and $H_{ij} \in I^2$. □

The $\mathfrak{sl}_2$ grading. Let $N \neq 0$ be a nilpotent linear vector field on $\mathbb{C}^n$. N acts on linear forms as a nilpotent operator; choose a Jordan basis $y_{b,1}, \ldots, y_{b,s_b}$ (blocks b) with $N(y_{b,i}) = y_{b,i+1}$, $N(y_{b,s_b}) = 0$, and use it as coordinates. There are linear vector fields E, H with

$$[H, E] = 2E, \qquad [H, N] = -2N, \qquad [E, N] = H, \qquad H(y_{b,i}) = (s_b + 1 - 2i)\,y_{b,i},$$

that is, (E, H, N) is an $\mathfrak{sl}_2$-triple (this is the Jacobson–Morozov theorem [Mo42], [Ja51]). Here $\mathfrak{sl}_2$ is the Lie algebra of complex 2×2 matrices with trace 0; its basis $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ satisfies the same relations, so $e \mapsto E$, $h \mapsto H$, $f \mapsto N$ is an action of $\mathfrak{sl}_2$ on polynomials. We need it for (S1) and (S2) below, which are facts about finite-dimensional $\mathfrak{sl}_2$-modules. We call $s_b + 1 - 2i$ the H-weight of $y_{b,i}$ and extend it additively to monomials. Every space of homogeneous polynomials of a fixed ordinary degree is a finite-dimensional $\mathfrak{sl}_2$-module, and N lowers the H-weight by 2. Two standard facts:

(S1) A nonzero H-homogeneous polynomial f with $N(f) = 0$ has H-weight ≤ 0.

(S2) For any polynomial G, the H-weight-0 part of $N(G)$, if it is killed by N, is 0.

Lemma 5.2. *Let $F \in \mathfrak{m}^3$ have an isolated critical point at 0, and let $N \neq 0$ be a nilpotent linear vector field. It is not possible that $N(F) = h$ where, in some linear coordinates, h contains only coordinates not moved by N (coordinates x_i with $N(x_i) = 0$).*

Proof. Use the Jordan coordinates. The linear forms ℓ with $N(\ell) = 0$ are the combinations of the y_{b,s_b}, so h is a polynomial in the y_{b,s_b}, and $N(h) = 0$.

Step 1: F has H-weights ≤ 1. Split $h = \sum_\mu h_\mu$ and $F = \sum_\mu F_\mu$ into H-homogeneous parts. Each h_μ is killed by N, so $\mu \leq 0$ by (S1), and $h_0 = 0$ by (S2). For $\mu \geq 2$, $N(F_\mu) = h_{\mu-2} = 0$, and F_μ has positive weight, so $F_\mu = 0$ by (S1).

Step 2: a critical line. Let $s \geq 2$ be the largest block size, b_0 a block of size s, $U = y_{b_0,1}$ (of H-weight $s-1$), and L the U-axis. A monomial of F contributes to a first derivative of F at points of L only if it is U^a or $U^a x$ with $x \neq U$ a coordinate; as $F \in \mathfrak{m}^3$, only U^a with $a \geq 3$ and $U^a x$ with $a \geq 2$ occur. Every coordinate has H-weight $\geq -(s-1)$, so these monomials have weight $\geq 3(s-1)$, resp. $\geq s-1$.

If $s \geq 3$, all these weights are ≥ 2, so by Step 1 none of these monomials occurs, and $\nabla F = 0$ on L.

If $s = 2$, the only remaining candidates are $U^2 y$ with y of weight -1, the second coordinate of a block of size 2. Write $t_1, \ldots, t_p$ for the first and $b_1, \ldots, b_p$ for the second coordinates of the blocks of size 2 ($t_1 = U$), and z for the coordinates of the blocks of size 1; then $N = \sum_i b_i \partial_{t_i}$ lowers the degree in t by 1, and $h \in \mathbb{C}[b, z]$. Let $F^{(2)}$ be the part of F of degree 2 in t; $N(F^{(2)})$ is the part of h of degree 1 in t, which is 0. The matrix $\mathcal{H} = (\partial_{t_i} \partial_{t_l} F^{(2)})$ has entries in $\mathbb{C}[b, z]$ and $\sum_i b_i \mathcal{H}_{il} = \partial_{t_l} N(F^{(2)}) = 0$. As $b_1, \ldots, b_p$ is a regular sequence in $\mathbb{C}[b, z]$, Lemma 5.1 gives $\mathcal{H}_{11} \in (b)^2$. The coefficient of U^2 in F is $\frac{1}{2}\mathcal{H}_{11}$, so F has no monomial $U^2 b_j$. Again $\nabla F = 0$ on L, contradicting isolatedness. □

6 Proof of Theorem 1.1

Notation. $S = \{m_1 w_1 + \cdots + m_n w_n : \ m_i \in \mathbb{Z}_{\geq 0}\}$ is the set of weights of monomials, and $S_k = \{m_k w_k + \cdots + m_n w_n : \ m_i \in \mathbb{Z}_{\geq 0}\}$ the set of weights of monomials in $x_k, \ldots, x_n$. For $q > 0$ and $2 \leq k \leq n$,

$$I_{k,q} = \{\, j < k : \ w_j - w_k - q > 0, \ w_j - w_k - q \in S \,\},$$

the columns in which row k of C can have a nonconstant entry.

Theorem 6.1. *Let F be as in Section 1 and $q > 0$. Suppose that for every k with $2 \leq k \leq n$ at least one of the following holds:*

(a) $|I_{k,q}| \le 2$*;*

(b) $d - w_k - q \notin S_k$*;*

(c) for no $j \in I_{k,q}$ *is* $w_j - w_k - q$ *the weight of a monomial of ordinary degree* ≥ 2*.*

Then $\mathrm{Mil}(F)$ *has no nonzero derivation of weight* $-q$*.*

Proof. Suppose $C \neq 0$, let k be its first nonzero row, and let V, h be as in Section 3. By Corollary 4.3, row k has at least three nonzero entries, all nonconstant, so they lie in columns of $I_{k,q}$.

If (a) holds for this k, this is a contradiction.

If (b) holds, then $h = \sum_{j<k} C_{kj}F_j$, a polynomial in $x_k, \dots, x_n$ of weight $d - w_k - q \notin S_k$, is 0. This is a relation among the regular sequence $F_1, \dots, F_n$, so every C_{kj} lies in J (as in Lemma 3.1); C_{kj} has weight $< w_1 < d - w_1$, so $C_{kj} = 0$, a contradiction.

If (c) holds, every nonzero C_{kj} is a nonconstant weighted homogeneous polynomial whose weight is not that of a monomial of degree ≥ 2, so it is a linear form. Then V is a nonzero linear vector field of weight $-(w_k + q) < 0$, hence nilpotent, and $V(F) = h$ contains only $x_k, \dots, x_n$, which V does not move. Lemma 5.2 gives a contradiction.

So $C = 0$, and there is no nonzero derivation of weight $-q$ (Section 3). □

Different rows may use different conditions (Example 7.3).

Proof of Theorem 1.1. We check the hypothesis of Theorem 6.1 in four cases. The least positive element of S is w_n, and a monomial of ordinary degree ≥ 2 has weight $\ge 2w_n$.

Case $q > w_3 - 2w_n$. For $j \ge 3$ and every $k > j$, $0 < w_j - w_k - q \le w_3 - w_n - q < w_n$ is impossible for an element of S, so $I_{k,q} \subset \{1, 2\}$ and (a) holds for every k.

Case $q > w_1 - 3w_n$. For every $j < k$, $w_j - w_k - q \le w_1 - w_n - q < 2w_n$, so (c) holds for every k.

In the other two cases w_n does not divide $d - q$. Since $S_n = w_n\mathbb{Z}_{\ge 0}$, $d - w_n - q \in S_n$ would mean that w_n divides $d - q$; so (b) holds for $k = n$. For $k \le n - 1$ we have $w_k \ge w_{n-1}$.

Case $q > w_3 - w_{n-1} - w_n$. For $3 \le j < k \le n - 1$, $w_j - w_k - q \le w_3 - w_{n-1} - q < w_n$, so $j \notin I_{k,q}$; hence $I_{k,q} \subset \{1, 2\}$ and (a) holds.

Case $q > w_1 - w_{n-1} - 2w_n$. For $j < k \le n - 1$, $w_j - w_k - q \le w_1 - w_{n-1} - q < 2w_n$, so (c) holds.

The first two cases are $q > M$; the last two are $q > M - (w_{n-1} - w_n)$ with $w_n \nmid d - q$, since $w_3 - w_{n-1} - w_n = (w_3 - 2w_n) - (w_{n-1} - w_n)$ and $w_1 - w_{n-1} - 2w_n = (w_1 - 3w_n) - (w_{n-1} - w_n)$. If $w_3 \le 2w_n$ or $w_1 \le 3w_n$, then $M \le 0 < q$ for every $q \ge 1$. □

Comparison. If $w_1 \le 2w_n$, then $w_1 \le 3w_n$; so Theorem 1.1 contains the Main Theorem of [YZ16] and the case of equal weights. For $n \le 4$ the absence of negative derivations is known for all weights ([CXY95], [Ch95]). For $n \ge 5$ the cases $w_3 \le 2w_n$ and $w_1 \le 3w_n$ are not covered by the results listed in Section 2, except for homogeneous F and for $d > w_1^2 - w_1 + 1$ [CY26]. The condition of Theorem 1.1 and the condition $d > w_1^2 - w_1 + 1$ do not contain each other (Examples 7.1 and 7.2).

7 Examples

Example 7.1. Weights $(12, 6, 6, 5, 5)$, $d = 60$, $F = x_1^5 + x_2^{10} + x_3^{10} + x_4^{12} + x_5^{12}$. Here $w_1 = 12 > 2w_5$, F is not homogeneous, and $d = 60 \le w_1^2 - w_1 + 1 = 133$. Since $w_1 \le 3w_5$ (and $w_3 \le 2w_5$), $M \le 0$, and by Theorem 1.1 $\mathrm{Mil}(F)$ has no nonzero derivation of negative weight for every isolated F with these weights.

Example 7.2. Weights $(4,4,4,1,1)$, $d=16$, $F=x_1^4+x_2^4+x_3^4+x_4^{16}+x_5^{16}$. Here $d=16>w_1^2-w_1+1=13$, so [CY26] applies. On the other hand $M=\min(4-2,\,4-3)=1$ and $M-(w_4-w_5)=1$, so Theorem 1.1 does not cover $q=1$.

Example 7.3. Weights $(7,7,7,3,2)$, $d=42$, $F=x_1^6+x_2^6+x_3^6+x_4^{14}+x_5^{21}$; here $d=42\le w_1^2-w_1+1=43$. $M=\min(7-4,\,7-6)=1$, so $q\ge 2$ is covered by $q>M$. For $q=1$: $M-(w_4-w_5)=0<1$ and $w_5=2$ does not divide $d-q=41$, so Theorem 1.1 applies. In Theorem 6.1: row 5 satisfies (b) only ($39\notin 2\mathbb{Z}$), and row 4 satisfies (c) only (3 is not the weight of a monomial of degree ≥ 2). So $\mathrm{Mil}(F)$ has no nonzero derivation of negative weight for every isolated F with weights $(7,7,7,3,2)$ and even d.

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